Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production backend experience in Python and Go, building distributed systems, REST APIs, and data pipelines at scale. Strong in data structures, algorithms, testing, and system design.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, Go, Java, JavaScript, TypeScript, C/C++, SQL.

Web & Backend: React, Node.js, Django, Flask, FastAPI, REST APIs, Microservices, Distributed Systems.

Databases: PostgreSQL, MySQL, Redis, Elasticsearch.

Developer Tools: Git, GitLab CI/CD, Jenkins, AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Bash, SonarQube.

AI & Data: Gemini, Vertex AI, PyTorch, TensorFlow, scikit-learn, LangChain, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Built AI-powered agent pipeline in Python decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** baseline across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) in automated investigation workflows while preserving task-completion fidelity.

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Built backend automation framework in **Python (Boto3)** for Teamcenter X infrastructure, replacing legacy scripts and reducing deployment time by **40%** across distributed AWS environments.
- Diagnosed and resolved live production issues across **44 environments** in 4 months across Linux, EC2, and databases.
- Migrated **40+ production environments** to Kubernetes-based distributed architecture using ArgoCD, Helm, and GitLab CI/CD.
- Increased automated test coverage from **35% to 78%** as Quality Coach; mentored **5 software engineers**.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Built **Go + Terratest** testing framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [Python, PySpark, FastAPI, PostgreSQL, Redis] Jul 2026

- Built backend ML system ingesting and processing **2.3M-row** time-series dataset across **263 zones**; designed two-stage model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%** over baseline.
- Built RESTful API in Python/FastAPI serving real-time predictions, integrated with PostgreSQL and Redis caching, containerized with Docker for cloud deployment.

PolicyPilot – Policy Intelligence Portal [TypeScript, Node, Gemini, MCP] May 2026

- Designed **17-tool MCP server** exposing structured LLM tool-calling APIs for automated document extraction and evaluation, consumable by external AI agents.
- Architected evidence-grounded RAG pipeline with citation tracing and confidence-scoring for human-review triage, eliminating hallucination in high-stakes data.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).